

UNT CSE Recruitment Advertisement for Prospective Ph.D. Students

Dr. Tong Shu (<https://tongshu83.github.io>), an fourth-year assistant professor in the Department of Computer Science and Engineering (CSE) at the University of North Texas (UNT), **will provide graduate assistantship for 2 Ph.D. students who start in Fall 2026**. Dr. Tong Shu received her B.S. degree from Peking University and her Ph.D. degree from New Jersey Institute of Technology, and **worked as a postdoc funded by the father of grid computing, Dr. Ian Foster, at Argonne National Laboratory (ANL) for three years. Her research interests include high-performance computing, quantum computing, parallel and distributed systems, and AI for sciences**. She published over 30 papers, such as top conference papers at SC, PPoPP, INFOCOM, and top journals at IEEE Transactions, and her student has published high-quality conference papers, such as IPDPS and Cluster. Also, **she obtained basic research grants of \$751,262 from the U.S. National Science Foundation, such as CAREER**.

Introduction to the University: The University of North Texas (UNT) in Denton (See Figure 1) is a national public university in the United States, and classified as an R1 institution, i.e., a doctoral university with very high research activity, by the Carnegie Classification of Institutions of Higher Education. In 2026, the UNT is globally ranked No. 714 and its CSE Department is nationally ranked No. 120 by U.S. News. Also, CSE at UNT is ranked No. 93 in terms of computer top conference papers by CSRankings. The UNT Denton campus (See Figure 2) has an area of 3.84 kilometers and is only 45 minute-driving away from downtown Dallas (See Figure 3), the ninth largest city in the U.S.. The UNT CSE has about thirty full-time tenured/tenure-track faculty members, more than two thousand undergraduate students, around six hundred master's students, and over one hundred PhD students. UNT College of Engineering (See Figure 4) is equipped with many modern devices, such as the Center for Agile and Adaptive Additive Manufacturing supported by \$10,000,000. Furthermore, UNT is close to the University of Texas at Dallas, Baylor University, Southern Methodist University, and the University of Texas at Arlington, which benefits research collaboration and spouse study.

Expectation to Prospective Students: Students are self-motivated and handy in system research on quantum computing (e.g., quantum machine learning and quantum simulation GPU superclusters) and have a good mathematics background, logical thinking, and perseverance.

Student Development: Offer (covering tuition and life expenses) and equipment (such as a workstation in the lab and access to GPU servers and large-scale clusters) will be provided for recruited PhD students. Since Dr. Shu maintains her long-term collaboration with ANL, her PhD students will have internship opportunities to work together with senior scientists at national labs.

How to Apply: Please email the resume (including education and work experience), all transcripts, TOEFL/IELTS and GRE score reports (required), and expertise (such as project experience and papers if applicable) to tong.shu@unt.edu .



Figure 1: Monument at University of North Texas



Figure 2: Main Campus of University of North Texas



Figure 3: Dallas Downtown



Figure 4: UNT College of Engineering